Saithan Panchaharan

saithanpanchaharan123@gmail.com | Personal Website | GitHub | LinkedIn

Education

University of Toronto Sep. 2024 – June 2029 *(Expected)*
BSc in Computer Science (Co-Op) *Toronto, ON*

- **Coursework**: *Software Engineering, Data Structures & Algorithms, Artificial Intelligence, Microprogramming, Assembly Programming, Operating Systems, Shell Programming, OOP, Statistics & Data Analysis*

Experience

Bellwoods Strategic Capital Sep. 2026 – Dec. 2026
Full-Stack Developer *Toronto, ON*

- Built **Google** and **Apple Sign-In** for **iOS** and **Android** with **React Native**, **Expo** and **Django REST Framework**, cutting sign-up from 7 fields to 1 tap (∼**86%** fewer inputs)
- Secured account linking against **pre-hijacking** via verified-email matching, validated by **20+ Django** unit tests across both sign-in providers
- Scoped **5** MVP user stories with the client and shipped **3** quest filters across **50+** quests, fixing a client-reported **Pixel 9** bug ahead of the **Google Play** release

Projects

Sportfolio | *Java, TypeScript, Hibernate, MLB Stats API* *Live Demo*

- Built a trading platform where users trade MLB/NFL players like stocks, using a **Spring Boot** pricing engine that blends 4 weighted signals into prices for **1,800+** players capped at **30%** daily movements
- Reduced daily repricing queries by **99%**, by batching **Spring Data JPA** queries into per-player maps via **Java Streams**
- Cut **Tank01** API calls to zero for historical repricing by archiving raw game stats in **PostgreSQL**, & cached player names, projections, & draft rankings for 24 hours
- Replaced a 1,900-parameter IN clause with a single date-bounded query, eliminating production database connection drops on **Railway** & keeping query size constant as the roster grows

LineTracker | *yfinance, Brevo, Google OAuth, JWT* *Live Demo*

- Built a full-stack platform with a **React** web app and **Flask** API that emails users the moment a stock price or sports betting line hits their target
- Reduced upstream API calls by **90%** by grouping active alerts by sport, market, & ticker, scaling **SharpAPI** & stock-price requests with distinct games & tickers, rather than alert count
- Checked every active alert every **90** seconds without breaching SharpAPI's free-tier limit of about 12 requests/minute, using **APScheduler**, **6**-second request spacing, & a **150**-second odds cache
- Shipped a native iOS app in **SwiftUI** on the same **Flask** and **PostgreSQL** backend, with **Keychain**-backed sessions & alerts to sync with the web app

Resume Genie | *Claude API, BeautifulSoup, pdfplumber, Docker* *Live Demo*

- Built a web app that tailors resumes to any job link by editing the original **LaTeX** or **Word** source in place
- Reduced model costs by **95%** through routed job checks, web searches, & tailoring to **Claude Haiku** over **Sonnet**
- Cut repeat job lookups to zero API calls by caching postings, pulling from **Greenhouse**, **Lever**, & **Ashby** first
- Reduced storage by **88%** & hosting to under **$1**/month by rebuilding PDFs on demand & moving the **Flask** backend to **Google Cloud Run**

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, C#, Go, Kotlin, Swift, SQL, Bash, HTML/CSS
Frameworks: React, Next.js, Angular, React Native, Node.js, Express.js, Spring Boot, Flask, FastAPI, Django, .NET, GraphQL, PyTorch, Pandas, Jest, SwiftUI
Tools: PostgreSQL, MySQL, MongoDB, Redis, Kafka, AWS, GCP, Azure, Docker, Kubernetes, Terraform, Linux, Git, GitHub Actions, CI/CD